

Topic: Minimum Mode Timing Diagrams

Q. Draw and explain the timing diagram for a Memory Read and Memory Write operation of the 8086 microprocessor in Minimum Mode. Detail the events occurring in each T-State.

[DEF] Definition to Remember

A Timing Diagram graphically represents the signals the CPU generates over time. In 8086, a basic bus cycle consists of 4 clock periods called T-States (T1, T2, T3, T4). T1 is for addressing, T2 for bus turnaround, T3 for data transfer, and T4 for cycle completion.

Minimum Mode Timing Diagrams

* T1: Address is generated and latched.

* T2

2. Memory Read Timing (Minimum Mode)

CPU fetches data from memory.

* T1

3. Memory Write Timing (Minimum Mode)

CPU sends data to memory.

* T1

[MUST-WRITE] Must-Write Points (for 10 marks)

1. A basic bus cycle consists of 4 clock periods: T1, T2, T3, T4.
2. T1: CPU outputs address on AD bus; pulses ALE high to latch the address.
3. T2: AD bus switches to data. Control signals (RD' or WR', and DEN') are asserted (pulled Low).
4. T3: Data is transferred (Read from or written to memory).

5. Wait States (T_W): Inserted between T₃ and T₄ if the READY pin is low (used for slow memory).

6. T₄: Control signals are deactivated (pulled High) and the cycle concludes.

7. DT/R' indicates direction (Low for Read, High for Write). M/IO' is High for Memory.

[RECALL] Quick Recall

Timing Diagram -> 4 T-States -> T₁ (Address out, ALE pulse) -> T₂ (RD/WR asserted, DEN asserted, Bus turnaround) -> T₃ (Data Transfer, Wait States if slow) -> T₄ (Signals deactivated)